Automated Reasoning and Learning - Introduction

Josef Urban

Czech Technical University in Prague

September 3, 2020

European Research Council
Established by the European Commission

Overview of Our Work

- Connections between two AI fields: Automated Reasoning (AR) and Machine Learning (ML)
- ML: apply various forms of *inductive reasoning* to large datasets to obtain the most plausible explanations, models and conjectures
- AR: apply various forms of *deductive reasoning* to prove that particular conjectures are correct
- AR includes *search* and *computation* - both rooted in *symbolic logic*
- Humans combine induction and deduction - let's teach computers too!
- *Typical problem:* How can learning help with logical reasoning?

Particular settings and topics

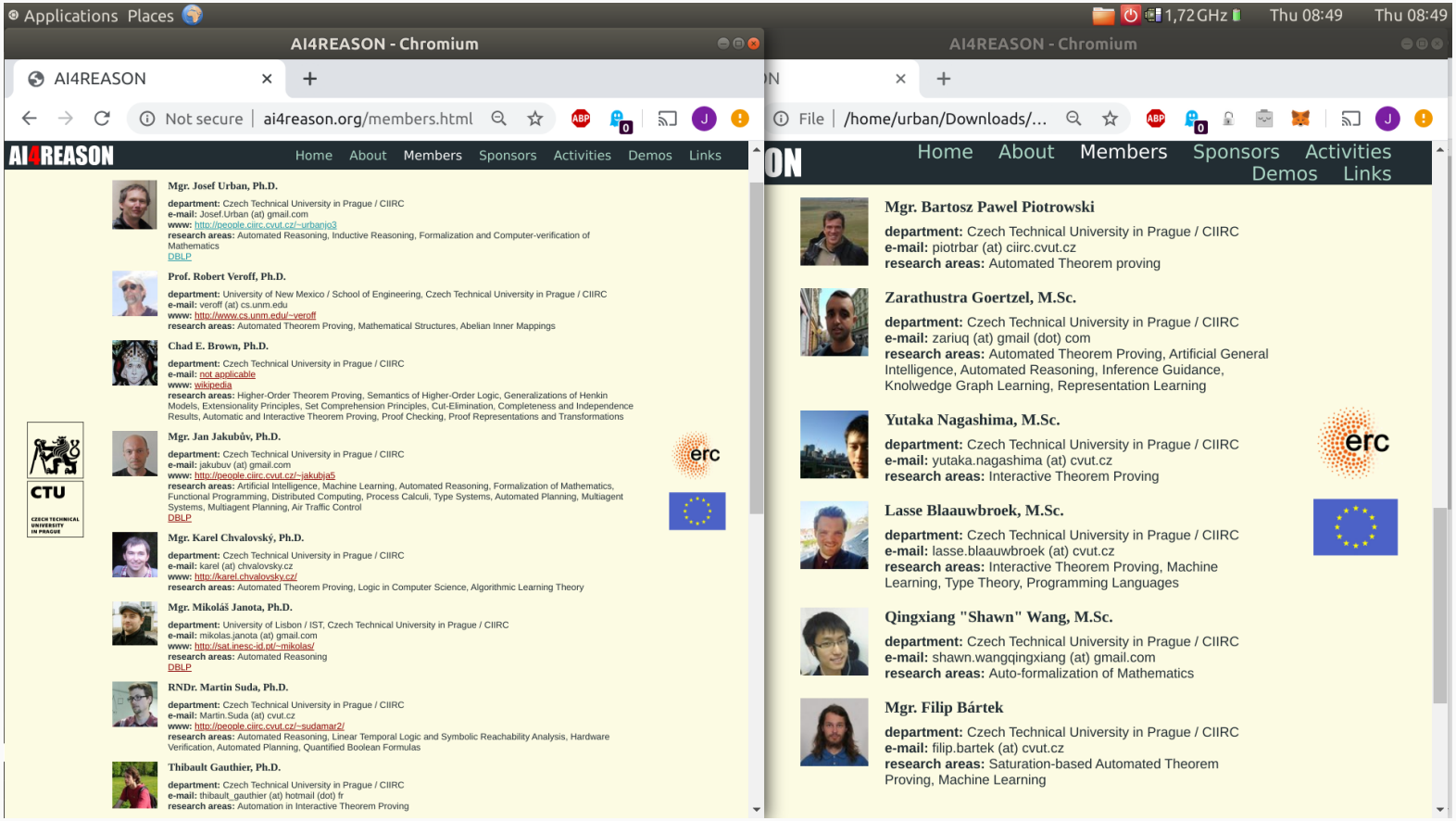
- ML and first-order logic (FOL), saturation-style theorem provers (ATPs)
- Higher-order logic (HOL), Set theory, formal proof assistants (ITPs)
- ML and reasoning in large theories, hammers for ITP, premise selection
- Symbolic vs statistical learning for theorem proving
- ML in tableau-style and tactical reasoning systems
- Learning in propositional logic (SAT), QBF, SMT, instantiation-based methods and model finding.
- Representations - how do we characterize reasoning data for learning?
- Conjecturing - how do we propose interesting mathematical conjectures?
- Feedback loops for proving and learning, reinforcement learning of ATP, positive/negative proof mining
- Alignment and translation between informal and formal corpora, automated formalization

Why Do This?

1. Practically Useful for Verification of Complex HW/SW and Math
   - Formal Proof of the Kepler Conjecture (2014 – Hales – 20k lemmas)
   - Formal Proof of the Feit-Thompson Theorem (2012 – Gonthier)
   - Verification of math textbooks and CS algorithms
   - Verification of compilers (CompCert)
   - Verification microkernels (seL4), HW chips (Intel)
   - Verification of cryptographic protocols, the OpenSSL stack (Amazon), etc.
2. Blue Sky AI Visions:
   - Get strong AI by learning/reasoning over large KBs of human thought?
   - Big formal theories: good semantic approximation of such thinking KBs?
   - Deep non-contradictory semantics – better than scanning books?
   - Gradually try learning math/science:
     - What are the components (inductive/deductive thinking)?
     - How to combine them together?

The Automated Reasoning Group at CIIRC

8 researchers/postdocs, 6 PhD students as of 2020 working in various topics



Automated Theorem Proving (ATP)

- try to infer conjecture C from axioms Ax: $Ax \vdash C$
- most classical methods proceed by refutation: $Ax \wedge \neg C \vdash \bot$
- $Ax \wedge \neg C$ are turned into *clauses*: universally quantified disjunctions of atomic formulas and their negations
- *skolemization* is used to remove existential quantifiers
- strongest methods: resolution (generalized modus ponens) on clauses:
- $\neg man(X) \vee mortal(X), man(socrates) \vdash mortal(socrates)$
- resolution/superposition (equational) provers generate inferences, looking for the contradiction (empty clause)
- main problem: combinatorial explosion especiialy over large theories - ML

What is Formal Mathematics?

- Developed thanks to the Leibniz/Russell/Frege/Hilbert/... program
- Mathematics put on formal logic foundations (*symbolic computation*)
- ... which btw. led also to the rise of computers (Turing/Church, 1930s)
- Formal math (1950/60s): combine formal foundations and the newly available computers
- Conceptually very simple:
- Write all your axioms and theorems so that computer understands them
- Write all your inference rules so that computer understands them
- Use the computer to check that your proofs follow the rules
- But in practice, it turns out not to be so simple
- Many approaches, still not mainstream, but big breakthroughs recently

Irrationality of $\sqrt{2}$ (informal text)

tiny proof from Hardy & Wright:

Theorem 43 (Pythagoras' theorem). $\sqrt{2}$ is irrational.
The traditional proof ascribed to Pythagoras runs as follows. If $\sqrt{2}$ is rational, then the equation

$$a^2 = 2b^2 \tag{4.3.1}$$

is soluble in integers a, b with $(a, b) = 1$. Hence a^2 is even, and therefore a is even. If $a = 2c$, then $4c^2 = 2b^2$, $2c^2 = b^2$, and b is also even, contrary to the hypothesis that $(a, b) = 1$. $\square$

Irrationality of $\sqrt{2}$ (Formal Proof Sketch)

exactly the same text in Mizar syntax:

```
theorem Th43: :: Pythagoras' theorem
  sqrt 2 is irrational
proof
  assume sqrt 2 is rational;
  consider a,b such that
4_3_1: a^2 = 2*b^2 and
    a,b are relative prime;
  a^2 is even;
  a is even;
  consider c such that a = 2*c;
  4*c^2 = 2*b^2;
  2*c^2 = b^2;
  b is even;
  thus contradiction;
end;
```

Irrationality of $\sqrt{2}$ in HOL Light

```
let SQRT_2_IRRATIONAL = prove
 (`~rational(sqrt(&2))`,
  SIMP_TAC[rational; real_abs; SQRT_POS_LE; REAL_POS] THEN
  REWRITE_TAC[NOT_EXISTS_THM] THEN REPEAT GEN_TAC THEN
  DISCH_THEN(CONJUNCTS_THEN2 ASSUME_TAC MP_TAC) THEN
  SUBGOAL_THEN `~((&p / &q) pow 2 = sqrt(&2) pow 2)`
    (fun th -> MESON_TAC[th]) THEN
  SIMP_TAC[SQRT_POW_2; REAL_POS; REAL_POW_DIV] THEN
  ASM_SIMP_TAC[REAL_EQ_LDIV_EQ; REAL_OF_NUM_LT; REAL_POW_LT;
               ARITH_RULE `0 < q <=> ~(q = 0)`] THEN
  ASM_MESON_TAC[NSQRT_2; REAL_OF_NUM_POW; REAL_OF_NUM_MUL; REAL_OF_NUM_EQ]);;
```

Today: Computers Checking Large Math Proofs

SCI NEWS

Researchers Find 40,000-Year-Old Figurative Paintings in Bornean Cave

HOME ASTRONOMY SPACE EXPLORATION ARCHAEOLOGY PALEONTOLOGY BIOLOGY PHYSICS MEDICINE GENETICS GEOLOGY MORE

Scientists Deliver Formal Proof of Famous Kepler Conjecture

Jun 16, 2017 by News Staff / Source

« Previous | Next »

Published in
Mathematics

Tagged as
Johannes Kepler
Kepler conjecture

Follow
You Might Like

Researchers Develop First-Ever 3D Numerical Model of Melting Snowflake

Researchers Develop Mathematical Model for How Innovations

An international team of mathematicians led by University of Pittsburgh Professor Thomas Hales has delivered a formal proof of the Kepler conjecture, a famous problem in discrete geometry. The team's paper is published in the journal *Forum of Mathematics, Pi*.

LATEST NEWS

SPHERE Captures Young Exoplanet Beta Pictoris b Orbiting around Its Star
Nov 13, 2018 | Astronomy

Mirarce eatoni: Newly-Discovered Cretaceous Bird Lived Among Dinosaurs, Was Strong Flier
Nov 13, 2018 | Paleontology

Juno Takes Closer Look at Jupiter's Magnificent, Swirling Clouds
Nov 13, 2018 | Space Exploration

Physicists Solve Structure of Unusually Complex Form of Nitrogen
Nov 13, 2018 | Physical Chemistry

Natural Compound Protects Hypertensive Rats against Heart Disease
Nov 13, 2018 | Medicine

Inventive Orangutans Make Hook Tools to Retrieve Food
Nov 12, 2018 | Biology

Researchers Find 40,000-Year-Old Figurative Paintings in Bornean Cave
Nov 12, 2018 | Archaeology

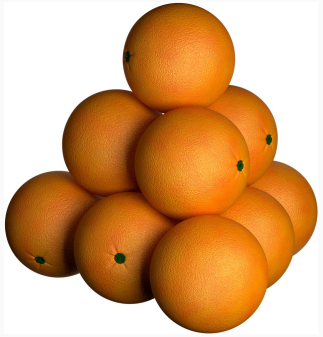
Hubble Sees Lensing Galaxy Cluster,

cdn.sci-news.com/images/enlarge3/image_4960e-Kepler-Conjecture.jpg

Big Example: The Flyspeck project

- Kepler conjecture (1611): The most compact way of stacking balls of the same size in space is a pyramid.

$$V = \frac{\pi}{\sqrt{18}} \approx 74\%$$



- Formal proof finished in 2014
- 20000 lemmas in geometry, analysis, graph theory
- All of it at `https://code.google.com/p/flyspeck/`
- All of it computer-understandable and verified in HOL Light:
- `polyhedron s /\ c face_of s ==> polyhedron c`
- However, this took 20 – 30 person-years!

Named Theorems in the Mizar Library

FM - Chromium

FM

fm.uwb.edu.pl/mmlquery/fillin.php?filledfilename=mml-facts.mqt&argument=number+102

Mizar home, download
files: abstr, articles, bin, doc, emacs gabs, fmbibs, gabs (more)
semantic MML

MML Query (beta)

Template maker
Environment explanation

Mizar TWiki
MML Query server
Megrez services
Journals:
FM: MetaPRESS, server, proof-read, regeneration
MM&A (preparation)

Syntax: xml, html
Downloads

Mizar syntax, xml, txt

MML 5.25.1220
- most important facts (other collection)

- Birkhoff

The most important facts in MML (decode)

add description

See also Name carrying facts/theorems/definitions in MML

1	"Alexander\'s Lemma"	=> WAYBEL_7:31	VOTE
2	"All Primes (1 mod 4) Equal the Sum of Two Squares"	=> NAT_5:23	VOTE
3	"Axiom of Choice"	=> WELLORD2:18	VOTE
4	"Baire Category Theorem (Banach spaces)"	=> LOPBAN_5:3	VOTE
5	"Baire Category Theorem (Hausdorff spaces)"	=> NORMSP_2:10	VOTE
6	"Baire Category Theorem for Continuous Lattices"	=> WAYBEL12:39	VOTE
7	"Banach Fix Point Theorem for Compact Spaces"	=> ALI2:1	VOTE
8	"Banach-Steinhaus theorem (uniform boundedness)"	=> LOPBAN_5:7	VOTE
9	"Bertrand\'s Ballot Theorem"	=> BALLOT_1:28	VOTE
10	"Bertrand\'s postulate"	=> NAT_4:56	VOTE
11	"Bezout\'s Theorem"	=> NEWTON:67	VOTE
12	"Bing Theorem"	=> NAGATA_2:22	VOTE
13	"Binomial Theorem"	=> BINOM:25	VOTE
14	"Birkhoff Variety Theorem"	=> BIRKHOFF:sch 12	VOTE
15	"Bolzano theorem (intermediate value)"	=> TOPREAL5:8	VOTE
16	"Bolzano-Weierstrass Theorem (1 dimension)"	=> SEQ_4:40	VOTE
17	"Borsuk Theorem on Decomposition of Strong Deformation Retracts"	=> BORSUK_1:42	VOTE
18	"Borsuk-Ulam Theorem"	=> BORSUK_7:condreg 3	VOTE
19	"Boundary Points of Locally Euclidean Spaces"	=> MFOLD_0:2	VOTE
20	"Brouwer Fixed Point Theorem"	=> BROUWER:14	VOTE
21	"Brouwer Fixed Point Theorem for Disks on the Plane"	=> BROUWER:15	VOTE
22	"Brouwer Fixed Point Theorem for Intervals"	=> TREAL_1:24	VOTE
23	"Brown Theorem"	=> GCD_1:40	VOTE
24	"Cantor Theorem"	=> CARD_1:14	VOTE
25	"Cantor-Bernstein Theorem"	=> CARD_1:10	VOTE

Automatically found proofs over the Mizar Library

https://github.com/ai4reason/ATP_Proofs/blob/master/README.md

Theorem Proving Competitions - CASC

http://www.tptp.org/CASC/J10/

Applications Places 1,33 GHz Thu 09:32 Thu 09:32

Results - Chromium

Not secure | tptp.org/CASC/J10/WWWFiles/DivisionSummary1.html

Higher-order Theorems	Zipperpin 2.0	Satallax 3.4	Satallax 3.5	Vampire 4.5	Leo-III 1.5	CVC4 1.8	LEO-II 1.7.0
Solved/500	424/500	323/500	319/500	299/500	287/500	194/500	112/500
Solutions	424 84%	323 64%	319 63%	299 59%	287 57%	194 38%	111 22%

Typed First-order Theorems +*-/	Vampire 4.5	Vampire 4.4	CVC4 1.8
Solved/250	191/250	190/250	187/250
Solutions	191 76%	190 76%	187 74%

First-order Theorems	Vampire 4.5	Vampire 4.4	Enigma 0.5.1	E 2.5	CSE_E 1.2	iProver 3.3	GKC 0.5.1	CVC4 1.8	Zipperpin 2.0	Etableau 0.2	Prover9 1109a	CSE 1.3	leanCoP 2.2	Twee 2.2.1	PyRes 1.3	lazyCoP 0.1
Solved/500	429/500	416/500	401/500	351/500	316/500	312/500	289/500	275/500	237/500	162/500	146/500	124/500	111/500	68/500	26/500	94/500
Solutions	429 85%	416 83%	401 80%	351 70%	316 63%	312 62%	289 57%	275 55%	237 47%	162 32%	146 29%	124 24%	111 22%	68 13%	26 5%	0 0%

First-order Non-theorems	Vampire SAT-4.5	Vampire SAT-4.4	iProver SAT-3.3	CVC4 SAT-1.8	E FNT-2.5	PyRes 1.3
Solved/250	238/250	226/250	182/250	98/250	63/250	13/250
Solutions	238 95%	226 90%	182 72%	98 39%	63 25%	13 5%

Unit Equality CNF	E 2.5	Twee 2.2.1	E 2.4	Vampire 4.5	Etableau 0.2	GKC 0.5.1	iProver 3.3	lazyCoP 0.1
Solved/250	202/250	197/250	185/250	162/250	148/250	128/250	124/250	20/250
Solutions	202 80%	197 78%	185 74%	162 64%	148 59%	128 51%	124 49%	0 0%

Large Theory Batch Problems	MaLARea 0.9	E LTB-2.5	iProver LTB-3.3	Zipperpin LTB-2.0	Leo-III LTB-1.5	ATPBoost 1.0	GKC LTB-0.5.1	Leo-III LTB-1.4
Solved/10000	7054/10000	3393/10000	3164/10000	1699/10000	1413/10000	1237/10000	493/10000	134/10000
Solutions	7054 70%	3393 33%	3163 31%	1699 16%	1413 14%	1237 12%	493 4%	134 1%

0

Using Learning to Guide Theorem Proving

- **high-level**: pre-select lemmas from a large library, give them to ATPs
- **high-level**: pre-select a good ATP strategy/portfolio for a problem
- **high-level**: pre-select good *hints* for a problem, use them to guide ATPs
- **low-level**: guide every inference step of ATPs (tableau, superposition)
- **low-level**: guide every kernel step of LCF-style ITPs
- **mid-level**: guide application of tactics in ITPs
- **mid-level**: invent suitable ATP strategies for classes of problems
- **mid-level**: invent suitable conjectures for a problem
- **mid-level**: invent suitable concepts/models for problems/theories
- **proof sketches**: explore stronger/related theories to get proof ideas
- **theory exploration**: develop interesting theories by conjecturing/proving
- **feedback loops**: (dis)prove, learn from it, (dis)prove more, learn more, ...
- ...

Sample of Learning Approaches We Have Been Using

- **neural networks** (statistical ML) – backpropagation, deep learning, convolutional, recurrent, etc.
- **decision trees, random forests, gradient tree boosting** – find good classifying attributes (and/or their values); more explainable
- **support vector machines** – find a good classifying hyperplane, possibly after non-linear transformation of the data (*kernel methods*)
- **k-nearest neighbor** – find the k nearest neighbors to the query, combine their solutions
- **naive Bayes** – compute probabilities of outcomes assuming complete (naive) independence of characterizing features (just multiplying probabilities)
- **inductive logic programming** (symbolic ML) – generate logical explanation (program) from a set of ground clauses by generalization
- **genetic algorithms** – evolve large population by crossover and mutation
- combinations of statistical and symbolic approaches (probabilistic grammars, semantic features, ...)
- supervised, unsupervised, reinforcement learning (actions, explore/exploit, cumulative reward)

Summary of Features Used

- From syntactic to more semantic:
- Constant and function symbols
- Walks in the term graph
- Walks in clauses with polarity and variables/skolems unified
- Subterms, de Bruijn normalized
- Subterms, all variables unified
- Matching terms, no generalizations
- terms and (some of) their generalizations
- Substitution tree nodes
- All unifying terms
- Evaluation in a large set of (finite) models
- Proximity to finished (sub)proofs
- LSI/PCA combinations of above
- Neural embeddings of above

Analogies with Molecular Interactions?

- atom ==> symbol
- molecule ==> term or formula
- many molecules interacting ==> proof search state (many formulas interacting)
- designing a molecule from many others ==> finding inferences of many formulas to infer a conjecture
- Our databases can have hundreds of thousands to milions of formulas
- Millions of proofs to learn from
- More proofs are easy to synthesize
- ML guidance: feed the proof state together with the possible actions to ML/NNs, RL settings

Example: Statistical Guidance the Given Clause in E Prover

- the E proof state are two large heaps of clauses *processed*/*unprocessed*
- the processed/selected clauses interact together, resulting in more unprocessed ones
- the hard part: choose the next clause to process (infer with)
- learn on E's proof search traces, put classifier in E
- positive examples: given clauses used in the proof
- negative examples: given clauses not used in the proof
- ENIGMA: fast feature extraction followed by a linear/boosted/neural classifier

Feedback loop for ENIGMA on Mizar data

- ENIGMA: guide inferences with boosted trees or NNs
- Done on 57880 Mizar problems very recently
- The proof search generates thousands to milions of formulas
- Ultimately a 70% improvement over the original strategy

	$\mathcal{S}$	$\mathcal{S} \odot \mathcal{M}_9^0$	$\mathcal{S} \oplus \mathcal{M}_9^0$	$\mathcal{S} \odot \mathcal{M}_9^1$	$\mathcal{S} \oplus \mathcal{M}_9^1$	$\mathcal{S} \odot \mathcal{M}_9^2$	$\mathcal{S} \oplus \mathcal{M}_9^2$	$\mathcal{S} \odot \mathcal{M}_9^3$	$\mathcal{S} \oplus \mathcal{M}_9^3$
solved	14933	16574	20366	21564	22839	22413	23467	22910	23753
$\mathcal{S}\%$	+0%	+10.5%	+35.8%	+43.8%	+52.3%	+49.4%	+56.5%	+52.8%	+58.4
$\mathcal{S}+$	+0	+4364	+6215	+7774	+8414	+8407	+8964	+8822	+9274
$\mathcal{S}-$	-0	-2723	-782	-1143	-508	-927	-430	-845	-454

	$\mathcal{S} \odot \mathcal{M}_{12}^3$	$\mathcal{S} \oplus \mathcal{M}_{12}^3$	$\mathcal{S} \odot \mathcal{M}_{16}^3$	$\mathcal{S} \oplus \mathcal{M}_{16}^3$
solved	24159	24701	25100	25397
$\mathcal{S}\%$	+61.1%	+64.8%	+68.0%	+70.0%
$\mathcal{S}+$	+9761	+10063	+10476	+10647
$\mathcal{S}-$	-535	-295	-309	-183

ENIGMA Proof Example – Knaster

```
theorem Th21:
  ex a st a is_a_fixpoint_of f
proof
  set H = {h where h is Element of L: h [= f.h};
  set fH = {f.h where h is Element of L: h [= f.h};
  set uH = "\/"(H, L);
  set fuH = "\/"(fH, L);
  take uH;
  now
    let fh be Element of L;
    assume fh in fH;
    then consider h being Element of L such that
A1: fh = f.h and
A2: h [= f.h;
    h in H by A2;
    then h [= uH by LATTICE3:38;
    hence fh [= f.uH by A1,QUANTAL1:def 12;
  end;
  then fH is_less_than f.uH by LATTICE3:def 17;
  then
A3: fuH [= f.uH by LATTICE3:def 21;
  now
    let a be Element of L;
    assume a in H;
    then consider h being Element of L such that
A4: a = h & h [= f.h;
    reconsider fh = f.h as Element of L;
    take fh;
    thus a [= fh & fh in fH by A4;
  end;
  then uH [= fuH by LATTICE3:47;
  then
A5: uH [= f.uH by A3,LATTICES:7;
  then f.uH [= f.(f.uH) by QUANTAL1:def 12;
  then f.uH in H;
  then f.uH [= uH by LATTICE3:38;
  hence uH = f.uH by A5,LATTICES:8;
end;
```

Statistical Guidance of Connection Tableau

- learn guidance of every clausal inference in connection tableau (leanCoP)
- set of first-order clauses, *extension* and *reduction* steps
- proof finished when all branches are closed
- a lot of nondeterminism, requires backtracking
- *Iterative deepening* used in leanCoP to ensure completeness
- good for learning – the tableau compactly represents the proof state

Clauses:

$c_1 : P(x)$

$c_2 : R(x, y) \vee \neg P(x) \vee Q(y)$

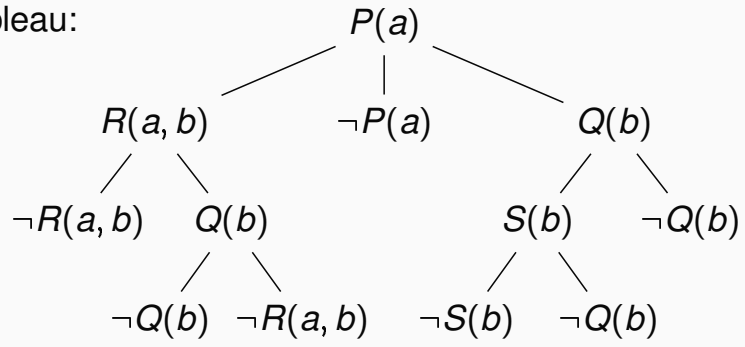
$c_3 : S(x) \vee \neg Q(b)$

$c_4 : \neg S(x) \vee \neg Q(x)$

$c_5 : \neg Q(x) \vee \neg R(a, x)$

$c_6 : \neg R(a, x) \vee Q(x)$

Closed Connection Tableau:



Statistical Guidance of Connection Tableau – rlCoP

- 2018: stronger learners via C interface to OCAML (boosted trees)
- remove iterative deepening, the prover can go arbitrarily deep
- added Monte-Carlo Tree Search (MCTS)
- MCTS search nodes are sequences of clause application
- a good heuristic to explore new vs exploit good nodes:

$$\frac{w_i}{n_i} + c \cdot p_i \cdot \sqrt{\frac{\ln N}{n_i}} \qquad \text{(UCT - Kocsis, Szepesvari 2006)}$$

- learning both *policy* (clause selection) and *value* (state evaluation)
- clauses represented not by names but also by features (generalize!)
- binary learning setting used: | proof state | clause features |
- mostly term walks of length 3 (trigrams), hashed into small integers
- many iterations of proving and learning

Statistical Guidance of Connection Tableau – rlCoP

- On 32k Mizar40 problems using 200k inference limit
- nonlearning CoPs:

System	leanCoP	bare prover	rlCoP no policy/value (UCT only)
Training problems proved	10438	4184	7348
Testing problems proved	**1143**	431	804
Total problems proved	11581	4615	8152

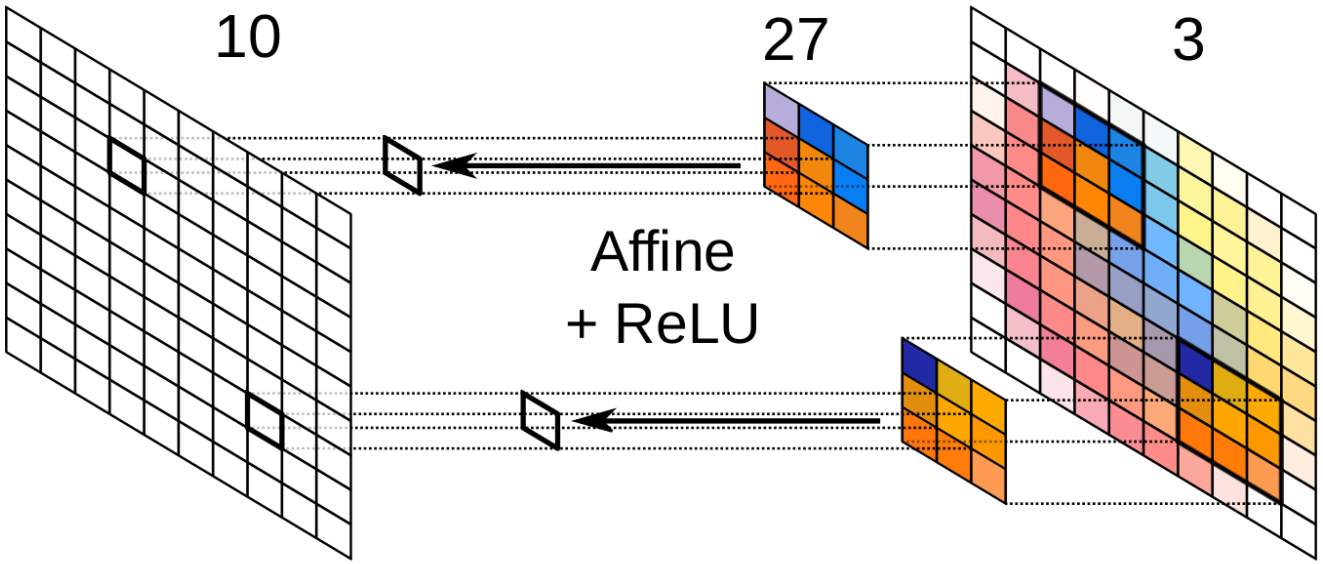
- rlCoP with policy/value after 5 proving/learning iters on the training data
- $1624/1143 = 42.1\%$ improvement over leanCoP on the testing problems

Iteration	1	2	3	4	5	6	7	8
Training proved	12325	13749	14155	14363	14403	14431	14342	**14498**
Testing proved	1354	1519	1566	1595	**1624**	1586	1582	1591

GNNs for Math - Mirek Olsak's architecture and presentation

Convolution

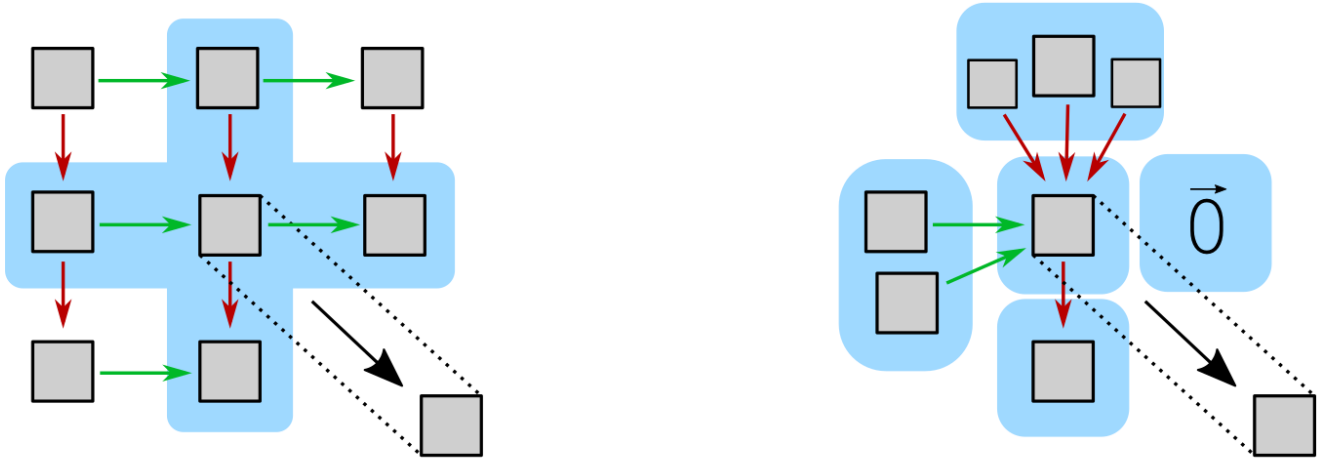
Basic layer for image processing: Picture → Picture



Applying the same basic layer locally around every pixel.

Convolution on a graph

Graph is a natural generalization of an image.



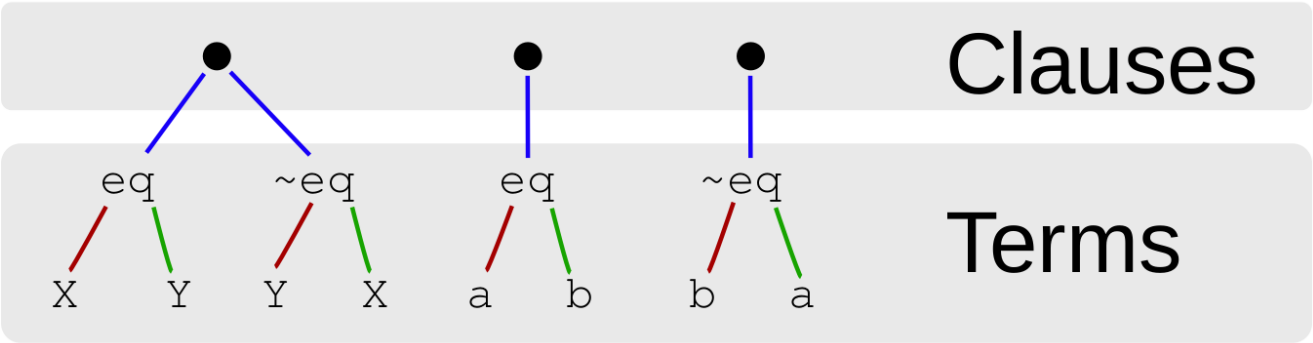
Evaluation of vertices by vectors $\rightarrow$ evaluation of the same vertices by vectors.

Multiple incoming edges of the same type are reduced via sum, maximum, mean, or their combination.

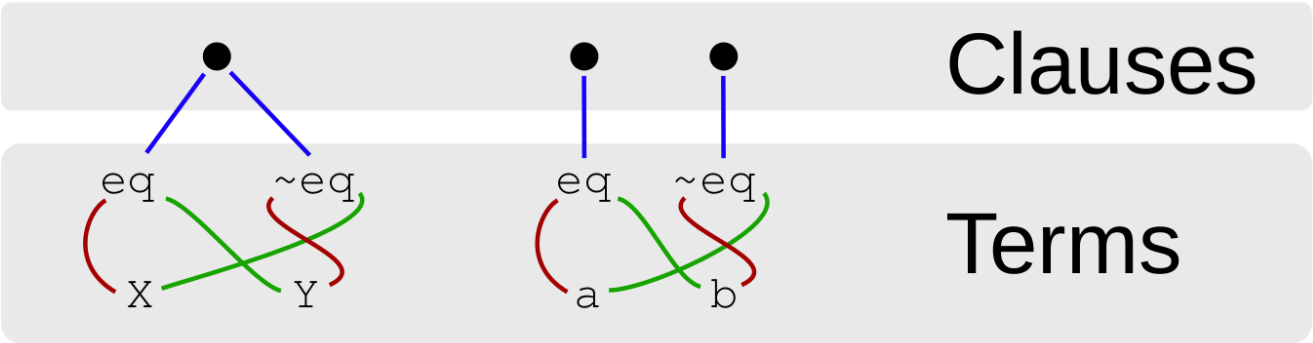
GNNs for Math

Mathematical formula

Tree



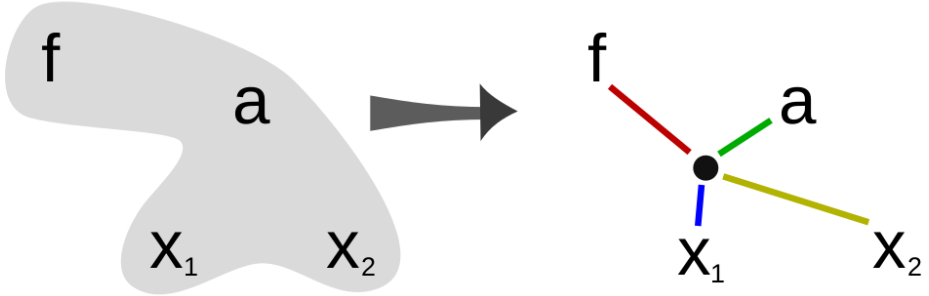
FormulaNet



Labels and arity

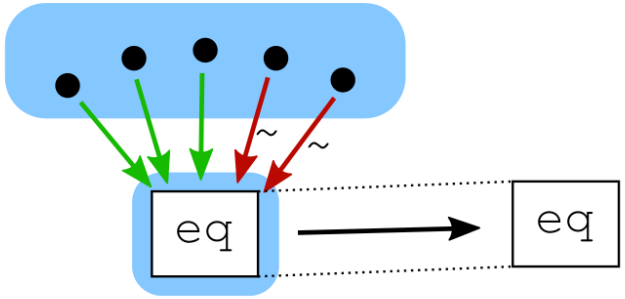
- Every label is represented by a node.
- Application $a = f(x_1, x_2, \ldots, x_n)$ is represented by a set of 4-ary hyperedges $(f, a, x_1, x_2), (f, a, x_2, x_3), \ldots, (f, a, x_{n-1}, x_n)$.

Hyperedges



Negation

- Boolean negation is represented by real negation (multiplication by -1)
- We apply this negation to outcoming edges, if required.
- How to keep incoming edges invariant under negation?
  - Reduction invariant under negation: average of min and max instead of max.
  - No biases. (Linear transformation, not affine one)
  - Odd activation function (tanh, not ReLU)
  - Labels initialized to zero.



GNNs for Math

Initialization

- Clauses:
  - three types in LeanCop (current goal, remaining goals, axioms),
  - two types in premise selection (conjecture, premises)
- Terms: constant embeddings for variables, literals, applications
- Label: Zero for relations, constant for functions

Premise selection

Dataset

- 32,524 queries extracted from Mizar
- Every query has balanced number of positive (useful) and negative results.
- Negative samples generated as the best scoring unuseful lemmas in k-nearest neighbors.

Results (on testing data)

	token sets	$\mathcal{M}_{\text{lin}}$	$\mathcal{M}_{\text{tree}}$	$\mathcal{M}_{\text{nn}}$	label-inv
Acc	75.75%	71.41%	77.98%	79.44%	80.36%
TPR	?	80.54%	83.35%	82.00%	84.46%
TNR	?	62.28%	72.60%	76.88%	76.25%

GNNs for Math

Label guessing

Data

- 1,517,692 occurences of 13,339 symbols

```
21.5%  def
17.3%  skolem
 2.0%  =
 1.7%  m1_subset_1
 1.2%  k1_zfmisc_1
 1.2%  r2_hidden
 1.0%  u1_struct_0
 0.9%  v1_funct_1
 0.9%  v1_xboole_0
 0.8%  v2_struct_0
       ...
```


More at `http://arxiv.org/abs/1911.12073` and `https://doi.org/10.1007/978-3-030-51054-1_29`

Transformers for Math - Conditioned Conjecturing

Taken from https://doi.org/10.1007/978-3-030-53518-6_24

```
for M, N being Cardinal holds

GPT-completes to:

( M = N iff M,N are_equipotent )
( M = N iff not 0 in M ) by ORDINAL3:8;
M *` N = N *` M
( M in N iff not M c= N )
the_rank_of M = the_rank_of N by Th77;
nextcard (Sum M) = M *` N
```

Proving neural completions

Applications Places 4,81 GHz Wed 14:42 Wed 14:42

emacs@dell

File Edit Options Buffers Tools Index Mizar Hide/Show Help

Save Undo

```
begin

for M, N being Cardinal holds card M c= M \/ N by XBOOLE_1:7,CARD_3:44,CARD_1:7,CARD_1:3; :: [ATP details]

for X, Y being finite set st not X is empty & X c= Y & card X = card Y holds X = Y by CARD_FIN:1; :: [ATP details]

for M, N being Cardinal holds
( M in N iff card M c= N ) by Unsolved; :: [ATP details]

for M, N being Cardinal holds
( M in N iff card M in N ) by CARD_3:44,CARD_1:9; :: [ATP details]

for M, N being Cardinal holds Sum (M --> N) = M *` N by CARD_2:65; :: [ATP details]


for M, N being Cardinal holds M /\ (union N) in N by Unsolved; :: [ATP details]


for M, N being Cardinal holds M *` N = N *` M by ATP-Unsolved; :: [ATP details]
```

-:--- **card_tst.miz** 3% L47 (Mizar Errors:2 hs Undo-Tree)

Wrote /home/urban/mizwrk/7.13.01_4.181.1147/tst8/card_tst.miz

Neural Autoformalization (Wang et al., 2018)

- Details in `https://doi.org/10.1007/978-3-319-96812-4_22` and `https://doi.org/10.1145/3372885.3373827`
- generate about 1M Latex - Mizar pairs based on Bancerek's work
- I.e. informal (Latex) to formal (Mizar)
- train neural seq-to-seq translation models (Luong – NMT)
- evaluate on about 100k examples
- many architectures tested, some work much better than others
- very important latest invention: *attention* in the seq-to-seq models

Neural Fun – Performance after Some Training

Rendered LaTeX	Suppose s_8 is convergent and s_7 is convergent . Then $\mathop{\rm lim}(s_8+s_7) = \mathop{\rm lim}\, s_8+\mathop{\rm lim}\, s_7$
Input LaTeX	`Suppose $ { s _ { 8 } } $ is convergent and $ { s _ { 7 } } $ is convergent . Then $ \mathop { \rm lim } ( { s _ { 8 } } { + } { s _ { 7 } } ) \mathrel { = } \mathop { \rm lim } { s _ { 8 } } { + } \mathop { \rm lim } { s _ { 7 } } $ .`
Correct	`seq1 is convergent & seq2 is convergent implies lim ( seq1 + seq2 ) = ( lim seq1 ) + ( lim seq2 ) ;`
Snapshot-1000	`x in dom f implies ( x * y ) * ( f | ( x | ( y | ( y | y ) ) ) ) = ( x | ( y | ( y | ( y | y ) ) ) ) ) ;`
Snapshot-2000	`seq is summable implies seq is summable ;`
Snapshot-3000	`seq is convergent & lim seq = 0c implies seq = seq ;`
Snapshot-4000	`seq is convergent & lim seq = lim seq implies seq1 + seq2 is convergent ;`
Snapshot-5000	`seq1 is convergent & lim seq2 = lim seq2 implies lim_inf seq1 = lim_inf seq2 ;`
Snapshot-6000	`seq is convergent & lim seq = lim seq implies seq1 + seq2 is convergent ;`
Snapshot-7000	`seq is convergent & seq9 is convergent implies lim ( seq + seq9 ) = ( lim seq ) + ( lim seq9 ) ;`

Neural Autoformalization results

Parameter	Final Test Perplexity	Final Test BLEU	Identical Statements (%)	Identical No-overlap (%)
128 Units	3.06	41.1	40121 (38.12%)	6458 (13.43%)
256 Units	1.59	64.2	63433 (60.27%)	19685 (40.92%)
512 Units	1.6	**67.9**	66361 (63.05%)	21506 (44.71%)
1024 Units	**1.51**	61.6	**69179 (65.73%)**	**22978 (47.77%)**
2048 Units	2.02	60	59637 (56.66%)	16284 (33.85%)